CAR

Conformalized Agentic Reasoning

Adaptive uncertainty-triggered verification for reliable multi-step LLM reasoning

Research Type Inference-time reliability framework	Primary Goal Reduce propagated reasoning errors while controlling verification cost
Core Model Open-weight LLM with accessible token scores	Core Mechanism Composite uncertainty + conformal calibration + selective verification

Project identity

This document is the standalone technical specification and research plan for CAR. It describes the intended system, mathematical design, data protocol, evaluation program, implementation roadmap, and reference literature. It does not depend on a review presentation.

Core thesis claim

A reasoning agent should not verify every step and should not trust every step. CAR estimates uncertainty at intermediate steps, calibrates that signal, and uses it to decide when external verification is worth the cost.

1. Executive Project Definition

CAR is an inference-time control framework for multi-step language-model reasoning. The agent produces structured intermediate claims. For each claim, CAR estimates uncertainty using token-level and semantic signals, maps that score through a calibration layer, and chooses between continuing, verifying, or abstaining. Verification uses external evidence or a deterministic tool when one exists.

Element	CAR definition
Problem	Early factual, logical, or computational mistakes become premises for later steps and produce propagated errors.
Input	A question that requires multiple dependent reasoning steps.
Core signal	Composite epistemic proxy based on token uncertainty, semantic divergence, and optional task-specific signals.
Calibration	Conformal calibration, with adaptive methods investigated for sequential or shifting conditions.
Action	CONTINUE, VERIFY, or ABSTAIN.
Verification	Evidence retrieval, a dedicated verifier, or a deterministic executor for math/code.
Output	A final answer whose reasoning path contains selectively verified steps.
Main hypothesis	Calibrated uncertainty produces a better accuracy-cost frontier than no verification, verify-everything, and heuristic uncertainty gates.

2. Research Problem

2.1 Reasoning snowballing

A multi-step agent is not a collection of independent predictions. Later claims condition on earlier claims. When an early claim is false and is treated as trusted state, later steps inherit the error. Recent 2026 work on multi-agent pipelines models this as an error-propagation process and reports that earlier verification boundaries preserve more opportunities to catch errors than end-of-pipeline checking.

Example chain

S1: retrieve a factual premise -> S2: derive an intermediate fact -> S3: compare two quantities -> S4: produce a conclusion. If S1 is false, S2-S4 can remain internally coherent while being globally wrong.

2.2 Why ordinary self-correction is insufficient

- Intrinsic self-correction depends on the same model signals that produced the first answer. Huang et al. report that LLMs struggle to self-correct reasoning without external feedback, and performance can degrade after self-correction.
- Self-verification experiments by Stechly et al. show substantial gains from sound external verification and performance collapse in some self-critique settings.
- Token probability is a model-distribution signal, not a truth certificate. A fluent false claim may still have low token entropy.
- Always-verifying is expensive and often unnecessary. The target is selective verification, not universal verification.

3. Research Objective and Questions

The project tests whether calibrated uncertainty is useful as a control signal for deciding when an agent should verify an intermediate reasoning step.

Primary research question

RQ1. Does conformally calibrated intermediate-step uncertainty detect incorrect reasoning early enough to reduce downstream error propagation?

Secondary questions

- RQ2. Does a composite lexical + semantic uncertainty score outperform a raw log-probability or entropy threshold?
- RQ3. Does adaptive calibration improve coverage or selective risk under sequential dependence or distribution shift relative to static split-conformal calibration?
- RQ4. How does verification frequency trade off against final accuracy and tool cost?
- RQ5. Does the same control principle transfer from factual QA to arithmetic reasoning and executable code?
- RQ6. Does verification at the earliest uncertain boundary reduce more downstream error than verification at the end?

Research hypothesis

H1: CAR will achieve a lower selective error rate at a fixed verification budget than heuristic uncertainty gating. H2: semantic uncertainty will add signal beyond token-level entropy. H3: early verification will produce a larger reduction in propagated errors than end-only verification. H4: a budget-aware threshold will improve the accuracy-cost Pareto frontier without violating the chosen empirical risk target.

4. Research Contribution

The project should be framed as a specific contribution, not as the first use of conformal prediction with agents. Prior work already applies conformal prediction to language models, agentic decision-making, and language-based planning. The CAR contribution is the focused combination of intermediate reasoning-step uncertainty, calibrated verification gating, sequential error-propagation analysis, and cost-aware evaluation.

Existing research line	What it establishes	CAR extension
Conformal language modeling	CP can provide statistical guarantees for sampled language-model outputs and can identify subsets of correct components.	Apply calibrated uncertainty as a control signal for intermediate reasoning verification.
Semantic entropy	Semantic disagreement across samples is useful for detecting hallucinations.	Fuse semantic divergence with token-level uncertainty at each reasoning step, then calibrate the combined score.
Agentic conformal decision-making	CP can constrain decisions or request help in agentic settings.	Move the decision target to reasoning-step verification and study downstream propagation.
Self-correction and external verification	Intrinsic self-correction is unreliable, while external verification helps.	Use uncertainty to decide when external verification is worth invoking.
Hallucination snowballing	Errors can become harder to detect after sequential agent handoffs.	Measure whether CAR catches errors before they become downstream state.

Core novelty target

A calibrated selective verification policy for intermediate reasoning states, evaluated against verification cost and downstream error propagation, with adaptive calibration treated as an explicit hypothesis rather than an assumed guarantee.

5. Statistical Foundation

5.1 Nonconformity and coverage

Let s(x,y) be a nonconformity score. A split-conformal procedure uses a held-out calibration set to estimate a quantile q_hat of the calibration scores. A prediction set or acceptance region is then defined by the score threshold. Under the usual exchangeability assumption, the resulting marginal coverage has a finite-sample guarantee.

Conceptual equation

C(x) = { y : s(x,y) <= q_hat } and P(Y in C(X)) >= 1 - alpha

Important CAR interpretation

The system should not describe 1 - alpha as model accuracy. It is a coverage or risk-control target associated with the chosen conformal construction and assumptions.

5.2 Sequential dependence and adaptive calibration

Reasoning steps are dependent because S_t conditions on prior state. Therefore, a standard exchangeability-based split-conformal guarantee should not be transferred to a dependent reasoning sequence without justification. CAR investigates adaptive conformal inference as the sequential layer. Gibbs and Candes show an online adaptive method that targets long-run coverage under distribution shift. For CAR, the adaptive update must be treated as a research component whose actual finite-sample or long-run guarantee depends on the precise update rule, feedback process, and assumptions.

Candidate adaptive update

q_hat(t+1) = q_hat(t) + gamma * (alpha - err_t)

Interpretation: if observed verification error rises above the target, the threshold should adapt toward more conservative gating. The exact direction, clipping, update schedule, and stability conditions must be specified in the implementation and evaluated empirically.

5.3 Do not claim more than the method proves

- A conformal layer does not make a language model truthful by itself.
- Adaptive conformal inference does not automatically provide a valid guarantee for arbitrary Markov reasoning chains.
- The calibration target must be defined precisely: accepted-step correctness, coverage, or a downstream task-risk measure.
- The final thesis should separate theoretical guarantees from empirical evidence.

6. Composite Epistemic Uncertainty

CAR uses multiple signals because no single uncertainty feature is a truth detector.

Signal	Definition	Strength	Failure mode
Token entropy	Entropy of next-token distribution over the generated claim/rationale.	Cheap and available from open models.	Low entropy does not imply truth.
Max surprisal	Maximum negative log probability token in the selected span.	Catches locally unusual tokens.	Sensitive to tokenization and stylistic effects.
Mean log probability	Average log probability over relevant tokens.	Simple sequence-level signal.	Length and syntax confounds.
Semantic divergence	Disagreement in meaning across sampled completions.	Directly targets epistemic variation.	Requires multiple samples and semantic clustering.
Verifier disagreement	Difference between independent verification signals.	Task-specific evidence.	Verifier itself can fail.

6.1 Semantic entropy

Farquhar et al. show that semantic entropy detects hallucination by grouping semantically equivalent answers rather than treating every lexical difference as uncertainty. CAR adapts the principle from whole-answer uncertainty to an intermediate reasoning step.

6.2 Candidate score

$$s_i = w1 * \text{TokenEntropy}_i + w2 * \text{SemanticDivergence}_i + w3 * \text{MaxSurprisal}_i + w4 * \text{TaskVerifierSignal}_i$$

The weights are fit on development data only. The calibration set is then used to map the scalar score into the decision rule. Test data stays locked.

7. Step Representation and Segmentation

CAR needs a stable unit of reasoning. Free-form paragraphs are too ambiguous for reproducible labels. The generator therefore emits structured JSON.

JSON schema

{ step_id, claim, rationale, dependency_ids, tool_request }

Segmentation rules

- One step corresponds to one atomic claim or computational transformation.
- dependency_ids explicitly identify earlier steps used as premises.
- Uncertainty is computed only on claim and rationale tokens, not JSON syntax.
- A step is accepted only if its dependency state is valid and the gate permits continuation.
- The system logs every decision, score, threshold, verification outcome, and downstream consequence.

8. Upgraded CAR Architecture

1. Query
Question requiring multi-step reasoning

2. Planner
Decompose into reasoning targets

3. Step Generator
Structured claim + rationale

4. Semantic Sampler
N alternative completions

5. Uncertainty Engine
Entropy + surprisal + semantic divergence

6. Conformal Engine
Calibrate score and threshold

7. Control Gate
CONTINUE / VERIFY / ABSTAIN

8. Verifier
Evidence or deterministic executor

9. State Update
Accept, rewrite, or stop

Control principle

The important architectural decision is that verification happens before the uncertain step is committed as trusted state. This changes the role of retrieval. Retrieval is not the default path for every step. It is a conditional intervention triggered by the calibrated gate.

9. CAR Control Policy

Reference pseudocode

```
INPUT: question Q, dev D, calibration C, test T, target alpha, budget B

weights <- fit_weights(D)
threshold <- calibrate(C, weights, alpha)
state <- planner(Q)

FOR each target in state.plan:
step <- generator(state)
features <- uncertainty(step)
score <- combine(features, weights)
decision <- gate(score, threshold, remaining_budget(B))

IF decision == CONTINUE:
accept(step)
observed_error <- 0

ELSE IF decision == VERIFY:
evidence <- retrieve(step, Q)
verdict <- verifier(step, evidence)

IF verdict == SUPPORTED:
accept(step)
observed_error <- 0
ELSE IF verdict == CONTRADICTED:
step <- revise(step, evidence)
accept(step)
observed_error <- 1
ELSE:
abstain_or_escalate(step)
observed_error <- 1

B <- B - 1

threshold <- adaptive_update(threshold, observed_error, alpha)

RETURN final_answer(state)
```

10. Verification Layer

Task family	Verification mechanism	Expected role
StrategyQA / factual QA	Retriever + evidence-grounded verifier	Check whether the step is supported or contradicted by evidence.
GSM8K	Calculator / symbolic arithmetic checker	Verify arithmetic and intermediate calculations.
HumanEval	Sandboxed Python test execution	Check executable functional correctness.
Medical QA	Retrieved authoritative evidence + cautious verifier	Stress test domain shift and evidence dependence.

Verifier design rule

The verifier should rely on an information source or deterministic procedure that is meaningfully different from the generator. A second LLM with the same parametric knowledge is not treated as a gold-standard verifier.

11. Benchmark and Data Protocol

Dataset	Role in CAR	Primary signal
StrategyQA	Primary multi-hop factual benchmark and calibration source.	Annotated decomposition plus evidence paragraphs.
GSM8K	Hard-logic propagation test.	Exact numerical answers and stepwise arithmetic.
HumanEval	Executable code generalization.	Unit tests and functional correctness.

Dataset	Role in CAR	Primary signal
MedQA	Domain-shift stress test for high-stakes factual QA.	Multiple-choice medical questions.

11.1 Data partitioning

- Development set: prompt design, score features, feature weights, and engineering choices.
- Calibration set: conformal threshold estimation only.
- Test set: never used for threshold selection, feature-weight tuning, or prompt selection.
- Cross-domain test: measure how calibration behaves when the test domain differs from the calibration domain.

11.2 Step labels

Generated chain-of-thought is not treated as ground truth. For StrategyQA, the benchmark decomposition and evidence provide the basis for step-level support checks. For GSM8K, mathematical execution provides deterministic checks. For HumanEval, unit tests provide the correctness signal. For MedQA, the benchmark answer plus retrieved evidence define the evaluation target.

12. Experimental Baselines

System	Verification policy	Why it is needed
CoT	Never verify	Lower-bound control.
Always Verify	Verify every step	Reliability upper-cost reference.
Entropy Gate	Verify when token entropy exceeds threshold	Tests raw lexical uncertainty.
Semantic Gate	Verify when semantic uncertainty exceeds threshold	Tests semantic uncertainty alone.
CoVe-style	Structured verification questions	Established verification workflow.
Reflexion	Self-reflection feedback	Intrinsic or feedback-based self-correction baseline.
Self-RAG	Adaptive retrieval + reflection tokens	Adaptive retrieval baseline.
Static CAR	Static conformal score threshold	Isolates adaptive calibration value.
CAR	Composite uncertainty + conformal gate + budget	Proposed system.

13. Core Experiments

Experiment A - Gate quality

For each generated step, measure whether the uncertainty score predicts step-level correctness. Report AUROC, AUPRC, calibration error, and selective risk.

Experiment B - Accuracy versus verification cost

Run all baselines under matched tool-call budgets. Plot final accuracy against verification calls per example. The intended result is a better Pareto frontier for CAR.

Experiment C - Propagation

Inject controlled errors into early steps. Compare no verification, end-only verification, and earliest-uncertain-step CAR. Record downstream step failure count and final-answer survival.

Experiment D - Adaptive calibration

Compare static conformal calibration with adaptive thresholding under simulated distribution shift and under sequential verification feedback. The objective is to test whether adaptive control maintains empirical risk better as the sequence context changes.

Experiment E - Cross-domain transfer

Calibrate on StrategyQA, then evaluate on MedQA and selected reasoning domains. Separately calibrate on the target domain. Compare coverage, selective risk, and verification rate.

Experiment F - Mechanistic analysis

Inject an incorrect premise at Step 1 and trace hidden-state or logit changes across later steps. The scientific goal is descriptive: measure whether internal representations shift as the corrupted premise becomes embedded in subsequent reasoning. Do not claim causal mechanism unless the intervention design supports a causal interpretation.

14. Evaluation Metrics

Metric	Definition	Use
Final accuracy	Correct final answer / examples	End-task performance.
False-safe rate	Incorrect steps accepted by the gate / incorrect steps presented to the gate	Primary safety-gate failure metric.
Step detection AUROC	Ranking quality of uncertainty score for wrong vs correct steps.	Measures signal quality.
Expected Calibration Error	Gap between confidence bins and empirical correctness.	Measures probability calibration.
Empirical coverage	Observed coverage of the conformal acceptance rule.	Checks target coverage.
Selective risk	Error rate among accepted outputs/steps.	Measures reliability under selective prediction.
Verification rate	Fraction of steps that trigger verification.	Measures intervention frequency.
Tool-call cost	Retriever + verifier calls per question.	Measures deployment cost.
Accuracy per tool call	Accuracy gain relative to verification cost.	Efficiency metric.
Latency	Wall-clock inference time per question.	Practical cost.

15. Ablation Program

- Remove semantic divergence and use token scores only.
- Remove token entropy and use semantic uncertainty only.
- Replace conformal calibration with a raw threshold.
- Replace adaptive calibration with static calibration.
- Remove the verification budget.
- Verify every step rather than selectively.
- Remove explicit dependency_ids and compare with free-form reasoning.
- Replace evidence-grounded verification with a same-model critic.

> The key ablation is static CAR vs adaptive CAR. The second key ablation is token-only vs composite uncertainty. These separate the value of calibration from the value of the uncertainty feature design.

16. Budget-Aware Control as a Constrained Decision Problem

Verification is a resource allocation problem. Each tool call has cost. The agent should seek acceptable risk subject to a finite budget.

State

State_t = {question, accepted_steps, current_step, uncertainty_score, threshold, remaining_budget, evidence_state}

Action

A_t in {CONTINUE, VERIFY, ABSTAIN}

Constraints

Expected selective risk <= target risk, and verification_calls <= B

Research use

The CMDP framing formalizes the accuracy-cost trade-off. Full reinforcement learning is optional. The core M.Tech implementation can use a deterministic budget-aware gate and treat CMDP optimization as the formal model for the policy.

17. Implementation Stack

Layer	Technology	Purpose
Model	Llama 3.1 8B or comparable open-weight model	Token scores and controlled generation.
Inference	PyTorch + Hugging Face Transformers	Logits, sampling, generation.
Agent graph	LangGraph or a lightweight custom state machine	Cyclic control and verification loop.
Retrieval	BM25 / dense retriever / vector DB	Evidence retrieval.
Verifier	LLM verifier with evidence, calculator, or Python sandbox	Independent correctness signal.
Conformal layer	Custom implementation	Research-controlled calibration logic.
Experiment tracking	Weights and Biases or MLflow	Runs, metrics, artifacts.
Visualization	Matplotlib / Plotly	Reliability diagrams, Pareto curves, error propagation plots.

Recommended compute

A single high-memory GPU is sufficient for the base experiment if sampling is controlled. Semantic sampling is the largest extra compute cost. Cache generated steps, hidden states where feasible, and retrieval results to reduce repeated inference.

18. Engineering and Safety Controls

- Run generated code only inside a sandbox with no network access and strict CPU, memory, and filesystem limits.
- Log evidence identifiers and verifier outputs so every verification decision is reproducible.
- Use a fixed test split and hash configuration files so test leakage is detectable.
- Treat retrieved documents as untrusted input and protect the verifier from prompt injection.
- Do not use external medical or financial outputs as production decisions. The project is an evaluation system.

19. Failure Modes and Mitigations

Failure	Mechanism	Mitigation
Confident hallucination	Model assigns high token probability to a false statement.	Semantic uncertainty + evidence-based verification.
Over-verification	Threshold is too conservative.	Budget-aware gating and Pareto analysis.
Under-verification	Threshold is too permissive.	Monitor false-safe rate and selective risk.
Calibration drift	Score distribution changes across domains or sequence states.	Adaptive calibration experiments and domain-transfer evaluation.
Verifier error	Verifier accepts a bad claim.	Ground verification in evidence or deterministic tools.
Label ambiguity	A reasoning step contains several claims.	Strict JSON schema and atomic step definitions.
Cost explosion	Semantic sampling multiplies inference.	Small N, caching, early stopping, and budget controls.
Leakage	Calibration or prompts are tuned on test examples.	Locked test set and separate configuration artifacts.

20. Mechanistic Propagation Study

The upgraded project includes a mechanistic analysis, but the analysis is deliberately scoped. It measures how an injected false premise changes internal signals across later steps.

Procedure

- Select a question with a known reasoning chain.
- Generate a clean trajectory and log hidden states and selected-token logits.
- Inject a controlled false premise at an early step.
- Continue generation without verification.
- Measure representational and logit divergence between clean and corrupted trajectories.
- Repeat with CAR verification enabled and compare downstream divergence.

Outputs

- Layer-wise divergence curves
- Step-wise confidence change
- Earliest detection point
- Downstream error count
- Effect of verification on divergence

Interpretation rule

This experiment supports a mechanistic account only to the extent that the intervention and measurements justify it. The baseline claim is descriptive: early errors alter subsequent model states and predictions.

21. Thesis Structure

Chapter	Content
1. Introduction	Problem, motivation, scope, research questions.
2. Background	LLM uncertainty, conformal prediction, semantic entropy, agent verification.
3. CAR Framework	Step representation, uncertainty score, calibration, control gate, verifier.
4. Experimental Method	Datasets, baselines, splits, metrics, statistical tests.

Chapter	Content
5. Results	Calibration, accuracy-cost frontier, propagation, cross-domain results.
6. Mechanistic Analysis	Controlled error injection and internal-state measurements.
7. Ablations and Failure Analysis	What works, what fails, sensitivity to assumptions.
8. Conclusion	Findings, limitations, future research.

22. Eight-Week Research Sprint

Weeks	Deliverable
1	Environment, Llama inference, structured step schema, logging.
2	Token uncertainty extraction and StrategyQA step pipeline.
3	Semantic sampling and composite uncertainty score.
4	Static conformal calibration and first reliability plots.
5	Verifier integration, Always-Verify, CoT, entropy baselines.
6	Adaptive calibration, budget-aware gate, Pareto experiments.
7	GSM8K, HumanEval, MedQA transfer and propagation injection.
8	Ablations, statistical tests, figures, thesis tables, final system packaging.

23. Project Success Criteria

- CAR produces reproducible uncertainty scores for intermediate steps.
- Calibration results are reported separately from test results.
- Empirical coverage and selective risk are measured, not asserted.
- CAR is compared with raw uncertainty thresholds and verify-everything baselines.
- The project shows the accuracy-cost trade-off rather than a single accuracy number.
- At least one experiment demonstrates early verification against end-only verification.
- The final report states precisely where the method does not generalize or where assumptions fail.

24. Expected Scientific Outcomes

There are three valuable outcomes even if the main hypothesis is not fully supported.

- Positive result: calibrated uncertainty improves selective verification efficiency.
- Mixed result: semantic uncertainty detects errors well, but adaptive calibration adds limited value.
- Negative result: token and semantic uncertainty fail to predict step correctness reliably. This itself yields a useful empirical result about the limits of uncertainty-based control.

25. Research References

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26. Reference-to-Component Map

CAR component	Primary references
Conformal output calibration	Quach et al.; Angelopoulos & Bates
Adaptive calibration	Gibbs & Candes
Semantic uncertainty	Farquhar et al.
Self-correction limits	Huang et al.; Stechly et al.
Verification baseline	Dhuliawala et al.
Agent reflection/retrieval	Shinn et al.; Asai et al.
Agentic conformal decision control	Wang et al.; Vishwakarma et al.; Si et al.
Snowballing motivation	Singh & Pawar
Factual benchmark	Geva et al. / StrategyQA
Math benchmark	Cobbe et al. / GSM8K
Code benchmark	Chen et al. / HumanEval

27. Final Technical Position

CAR is a selective verification architecture for multi-step LLM reasoning. Its research value comes from evaluating the full control loop: uncertainty estimation, calibration, verification timing, downstream propagation, and cost. The project should treat adaptive conformal inference as a hypothesis to validate under sequential dependence, not as an automatic guarantee imported from i.i.d. conformal prediction.

Core experimental statement

If CAR is successful, a model should spend verification calls where uncertainty and expected downstream loss are highest, while maintaining a measurable selective-risk target and producing a better accuracy-cost Pareto frontier than static or heuristic gates.

End of project deep-dive